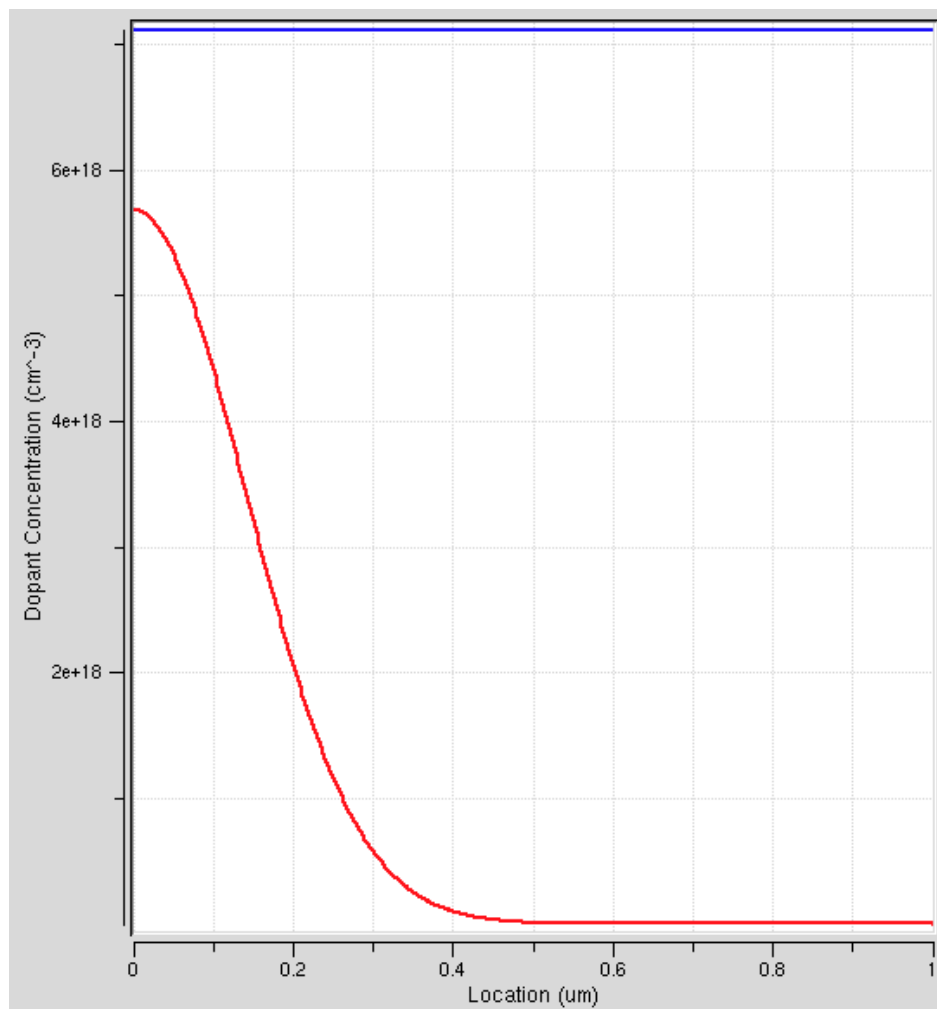


Rahul Bhadane

23104B0003

Experiment 2: Diffusion Process

Delta or Infinite Source:	delta
Time (minutes):	120
Temperature (C):	1000C
Bulk Length (um):	1
Concentration Dependence:	No dependence
choose one dopant:	P
Initial Dopant C (cm ⁻³):	1e20
Dose (cm ⁻²):	1e14



Delta or Infinite Source:

Time (minutes):

Temperature (C):

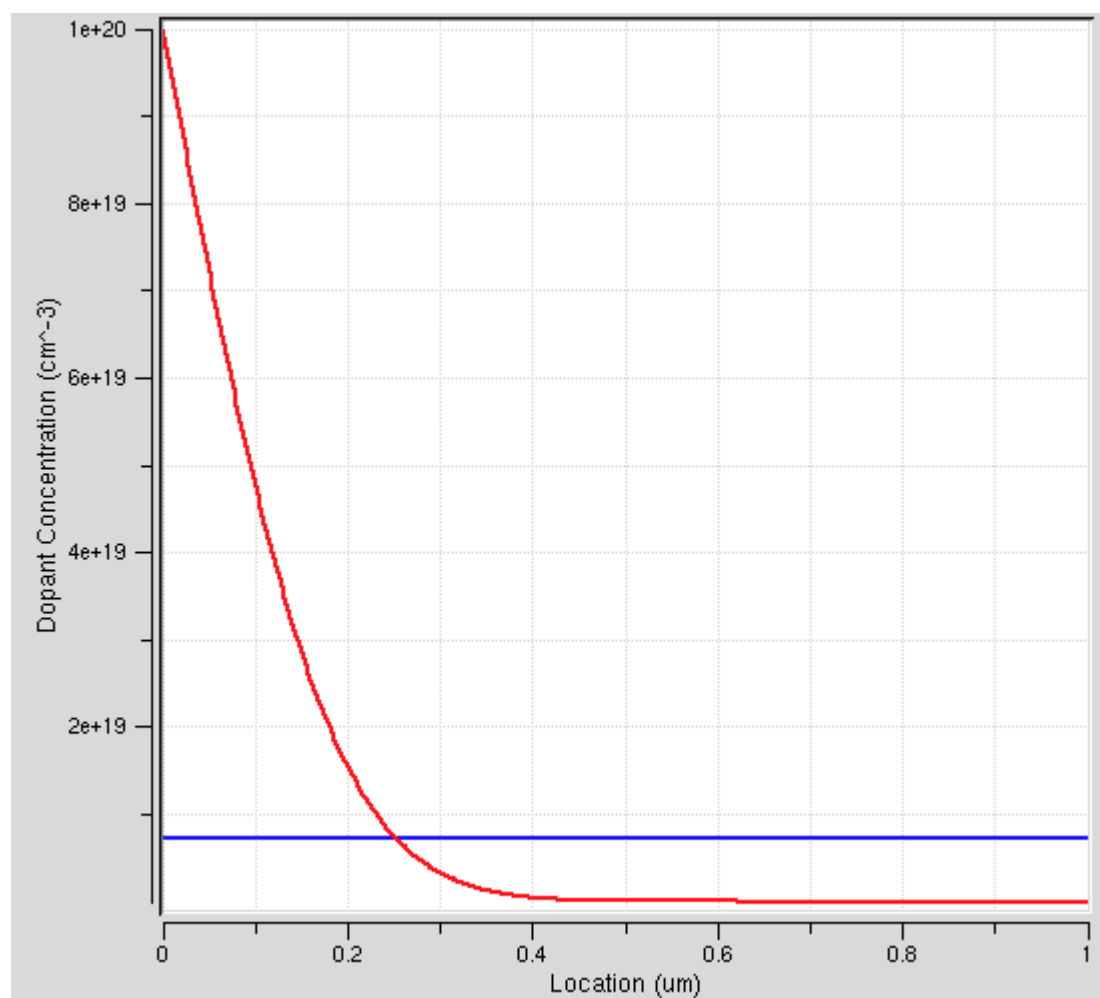
Bulk Length (um):

Concentration Dependence:

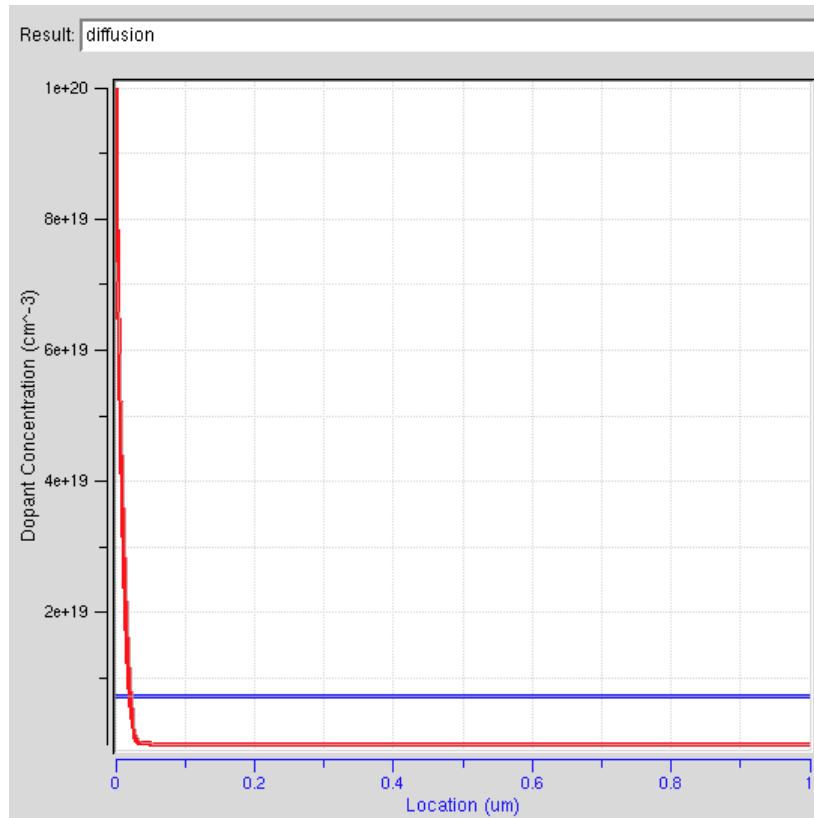
choose one dopant:

Initial Dopant C (cm⁻³):

Dose (cm⁻²):



Silicon Delta(pink) and infinite(red)



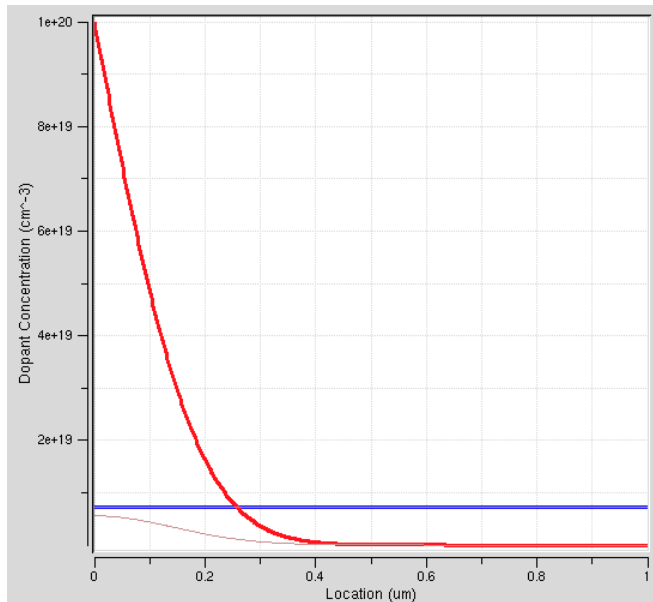
Analysis

The delta diffusion profile shows a sharp peak near the surface that decreases rapidly with depth, indicating a limited dopant supply. In contrast, the infinite source diffusion profile exhibits a higher and more gradual concentration gradient extending deeper into the silicon, as the dopant supply is continuously maintained at the surface.

Conclusion

Delta diffusion results in shallow and localized doping, while infinite source diffusion produces deeper and more uniform dopant penetration. This demonstrates the strong dependence of junction depth and concentration profile on the diffusion source type.

Boron



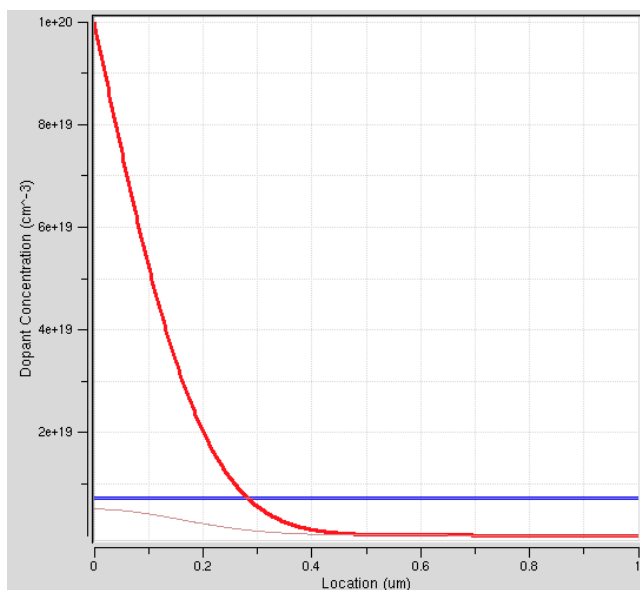
Analysis

The boron diffusion curve spreads relatively deeper into silicon compared to heavier dopants. The concentration decreases smoothly with depth, indicating high diffusivity.

Conclusion

Boron diffuses rapidly in silicon, making it suitable for forming deeper p-type regions. Its lighter atomic mass leads to higher diffusion rates compared to heavier dopants.

Indium



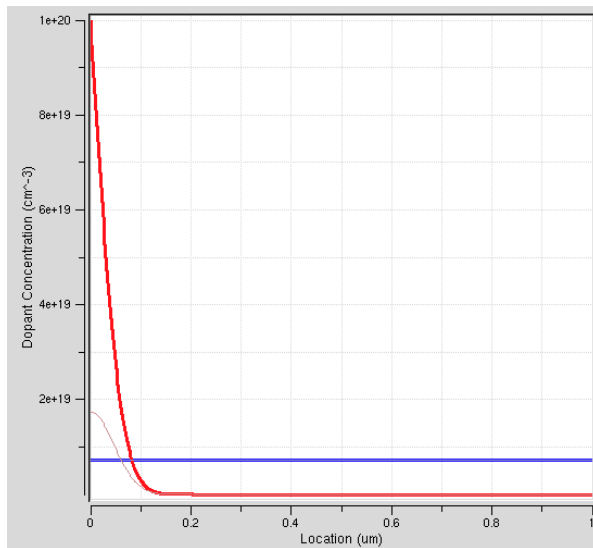
Analysis

The indium profile shows a steeper concentration drop with depth, indicating slower diffusion. The dopant remains more confined near the surface.

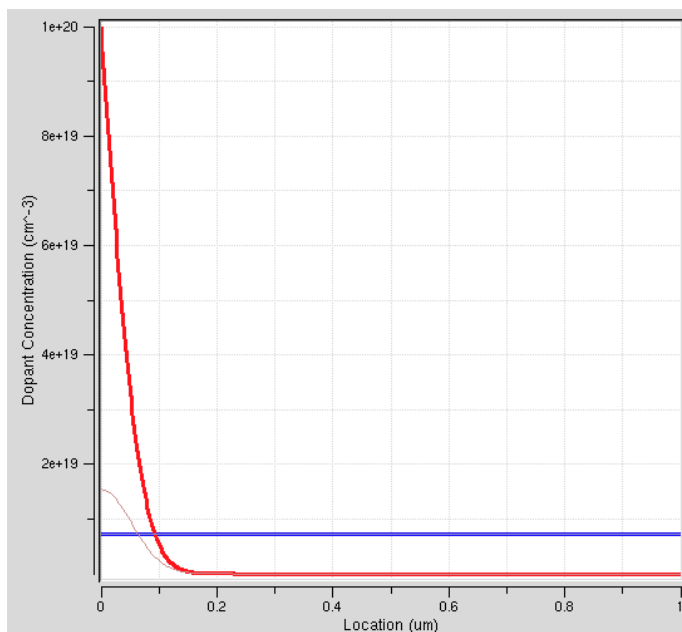
Conclusion

Indium has lower diffusivity in silicon, resulting in shallow junctions. This makes it suitable for applications requiring precise and localized p-type doping.

Arsenic



Antimony



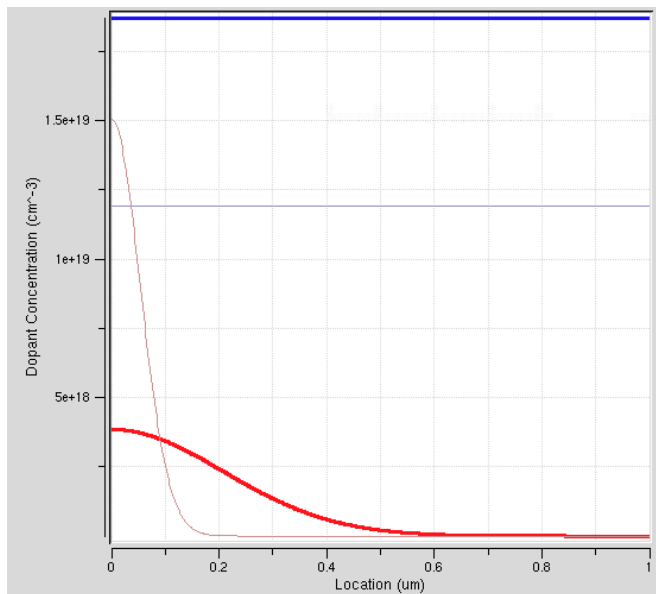
Analysis

Antimony shows the most confined diffusion profile among the dopants. The concentration falls very rapidly with depth.

Conclusion

Antimony has the lowest diffusivity in silicon, making it ideal for ultra-shallow and well-controlled n-type junction formation.

Delta temp vary silicon 1100(pink) and 1200(red)



Effect of Temperature Variation (1100 °C vs 1200 °C) in Silicon Diffusion

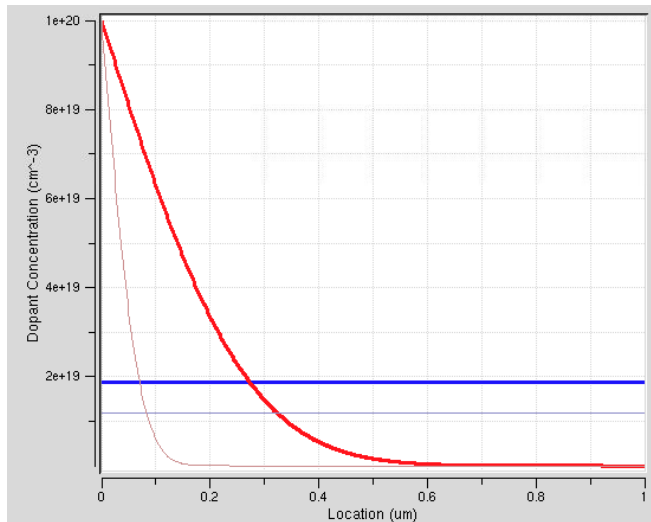
Analysis

At higher temperature (1200 °C), the diffusion profile spreads deeper into silicon with a smoother gradient compared to 1100 °C. Increased thermal energy enhances dopant mobility.

Conclusion

An increase in diffusion temperature significantly increases dopant penetration depth, confirming that diffusion is a thermally activated process.

Infinite



Effect of Temperature Variation in Infinite Source Diffusion

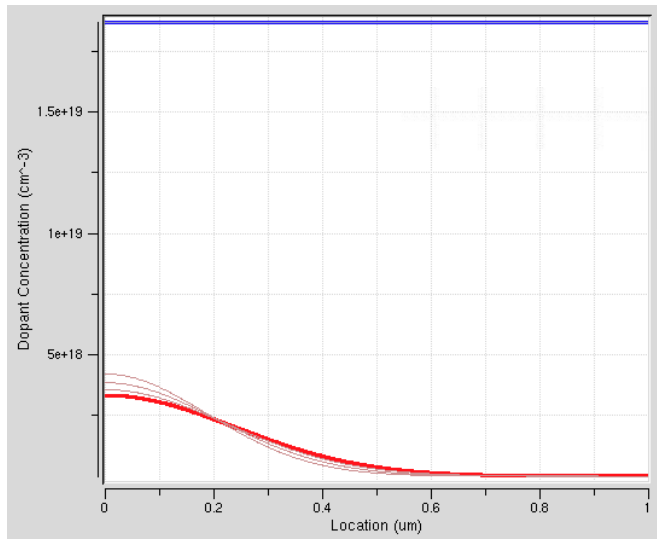
Analysis

In infinite source diffusion, increasing the diffusion temperature causes the dopant profile to extend deeper into the silicon substrate. While the **surface concentration remains constant** (due to the continuous dopant supply), higher temperature increases the dopant's diffusivity. As a result, the concentration gradient becomes smoother and the junction depth increases with temperature.

Conclusion

An increase in temperature significantly enhances dopant penetration depth in infinite source diffusion without altering surface concentration. This confirms that diffusion is a **thermally activated process**, where temperature primarily controls junction depth and profile spread.

Delta time vary 100 120 140 160 dark being 160



Effect of Time Variation (100, 120, 140, 160 units) in Delta Diffusion

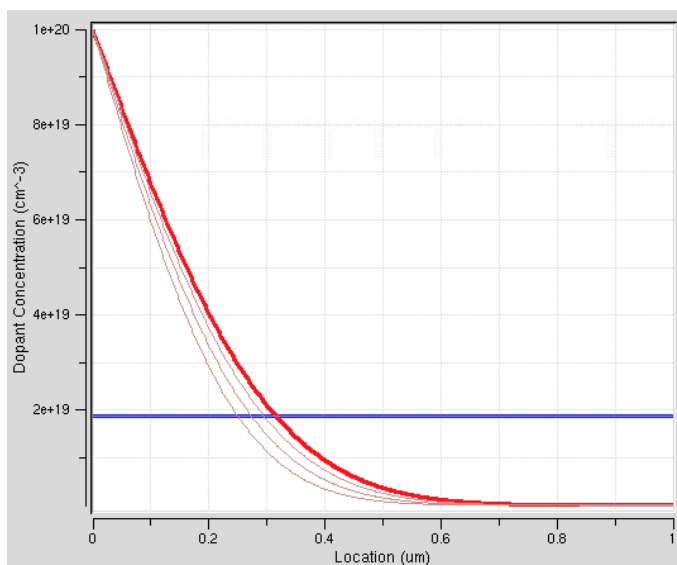
Analysis

As diffusion time increases, the dopant profile becomes broader and penetrates deeper into silicon. The darkest curve (maximum time) shows the widest spread.

Conclusion

Longer diffusion time increases junction depth and dopant spread, validating that diffusion depth is proportional to the square root of time.

Infinite



Infinite Source Diffusion with Time Variation

Analysis

With increasing time, the surface concentration remains nearly constant while the dopant penetrates deeper into the substrate.

Conclusion

Infinite source diffusion maintains a fixed surface concentration, and increased time primarily affects junction depth rather than peak concentration.

CONCLUSION

The diffusion characteristics in silicon are strongly influenced by the **type of dopant, diffusion source, temperature, and diffusion time**. Lighter dopants exhibit deeper and faster penetration, while heavier dopants result in shallow and more abrupt profiles. Delta diffusion produces localized and shallow doping due to limited dopant availability, whereas infinite source diffusion leads to deeper junctions with a constant surface concentration. Increasing temperature and diffusion time enhances dopant mobility, resulting in greater penetration depth and smoother concentration gradients. These observations confirm that diffusion is a **thermally activated process** and highlight the importance of parameter control for precise junction formation in semiconductor device fabrication.